

8ai	By conservation of energy, Loss in Electric PE = Gain in KE $q\Delta V = \frac{1}{2}mv^2 = \frac{p^2}{2m}$ $(1.60 \times 10^{-19})(130) = \frac{p^2}{2(9.11 \times 10^{-31})}$ $p = 6.1561 \times 10^{-24}$ $\lambda = \frac{h}{p}$ $= \frac{6.63 \times 10^{-34}}{6.1561 \times 10^{-24}}$ $= 1.08 \times 10^{-10} \text{ m}$	C1 C1 A1
aii	The separation of the silicon atoms is in the same order of magnitude as the de Broglie wavelength of the electrons. Hence effect of diffraction is appreciable/significant	A1
bi	For a wave, electron can <u>'collect' energy continuously</u> for a wave, electron will always be emitted / electron will be <u>emitted at all frequencies</u> after a <u>sufficiently long delay</u> to collect sufficient energy	B1 B1 B1
bii1	Either wavelength is longer than threshold wavelength or frequency is below the threshold frequency or photon energy is less than work function	A1
bii2	$hc / \lambda = \phi + E_k$ $(6.63 \times 10^{-34} \times 3.00 \times 10^8) / (240 \times 10^{-9}) = \phi + 4.44 \times 10^{-19}$ $\phi = 3.8 \times 10^{-19} \text{ J (allow } 3.9 \times 10^{-19} \text{ J)}$	C1 A1
biii1	photon energy larger so (maximum) kinetic energy is larger	M1 A1
biii2	fewer photons per unit time so (maximum) current is smaller	M1 A1
ci 1 2	arrow from -0.54 eV to -0.85 eV, labelled L arrow from -0.54 eV to -3.4 eV, labelled S (two correct arrows, but only one label – allow 2 marks) (two correct arrows, but no labels – allow 1 mark)	A1 A1
cii	$E = hc / \lambda$ $(3.4 - 0.54) \times 1.6 \times 10^{-19} = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / \lambda$ $\lambda = 4.35 \times 10^{-7} \text{ m}$	C1 A1
ciii	$-1.50 \rightarrow -3.4 = 1.9 \text{ eV}$ $-0.85 \rightarrow -3.4 = 2.55 \text{ eV (allow } 2.6 \text{ eV)}$ $-0.54 \rightarrow -3.4 = 2.86 \text{ eV (allow } 2.9 \text{ eV)}$ 3 correct energies – 2 marks 1 or 2 correct energies – allow 1 mark	A2
Total:		20